رواد مصر الرقمية

وزارة الاتصالات
وتكنولوجيا المعلومات

Project Title :VLAN and Inter-VLAN Routing Using FortiGate

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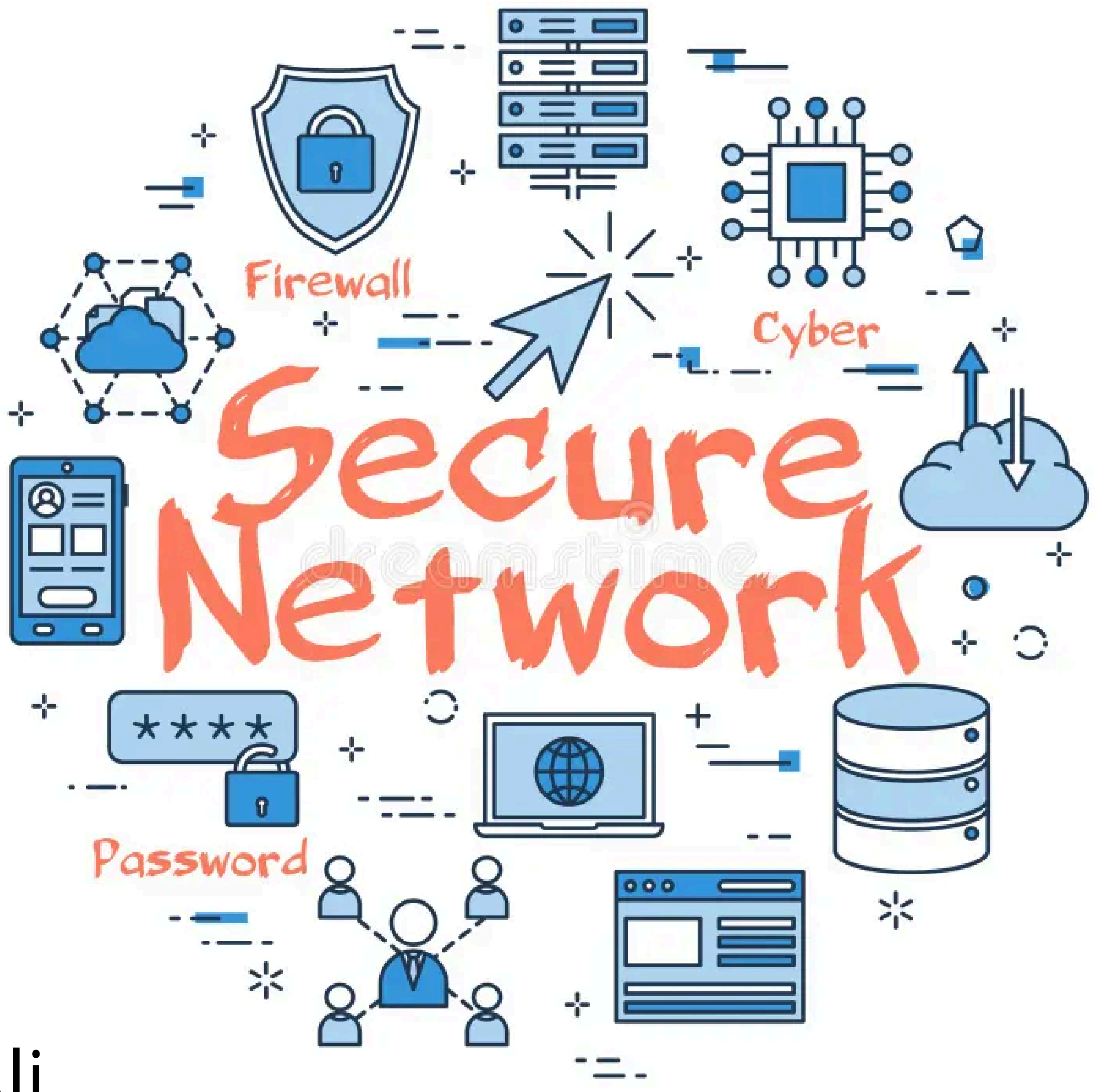

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TABLE OF CONTNTS

PROJECT OVERVIEW

- Implement VLANs on a switch
- Integrate FortiGate for routing
- Apply firewall policies
- Test communication between VLANs

FORTIGATE FIREWALL

Firewall Evolution

Transparent Firewall

- This firewall does not have an IP address
- Work as a layer 2 switch
- No NAT
- No routing
- Inspection of traffic up to layer 2

What is a Firewall

- A firewall is a network security device that monitors and filters incoming and outgoing network traffic based on an organization's previously established security policies.
- A firewall's main purpose is to allow non-threatening traffic in and to keep dangerous traffic out.

First & Second Generation (Traditional Firewall).

- Work at OSI Layer 3 & 4
- NAT
- Routing
- Traffic Inspection
- VPN
- No IPS
- Do not have application awareness

UTM (Unified Threat Management)

Benefits

- Work at OSI Layer 7
- All-in-one network security platforms.
- Firewall - VPN
- IPS-IDS

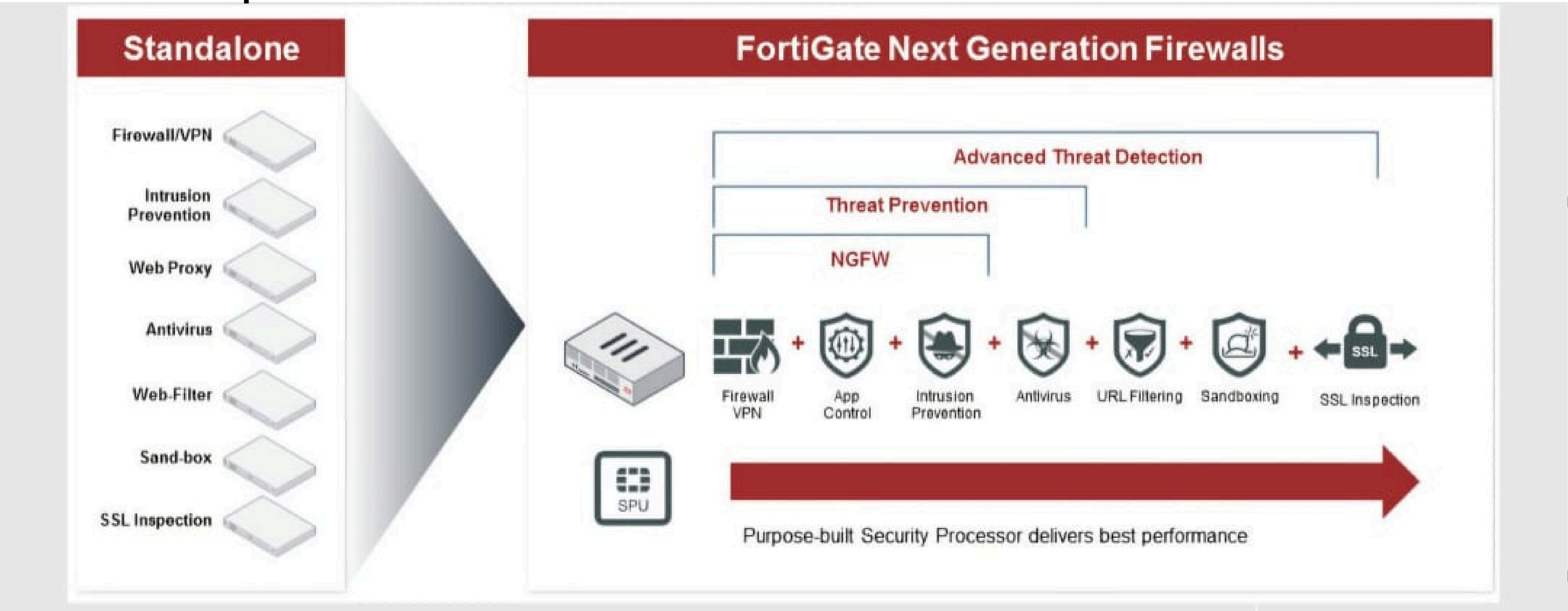
Antivirus-Antispyware

- Web Filtering-Mail Filtering

NGFW (Next Generation Firewall)

It is very similar to UTM, but with limited feature

That means NGFWs may include IPS, antivirus and malware prevention, application control, deep packet inspection and stateful firewalls, encryption, compression, QoS and other capabilities



VLANS

- What is VLAN?

A VLAN (Virtual LAN) allows devices on the same physical network to be grouped into separate virtual networks.
👉 Improves security, performance, and traffic management.

- Why do we need VLANs?

We use VLANs to separate departments within a company:

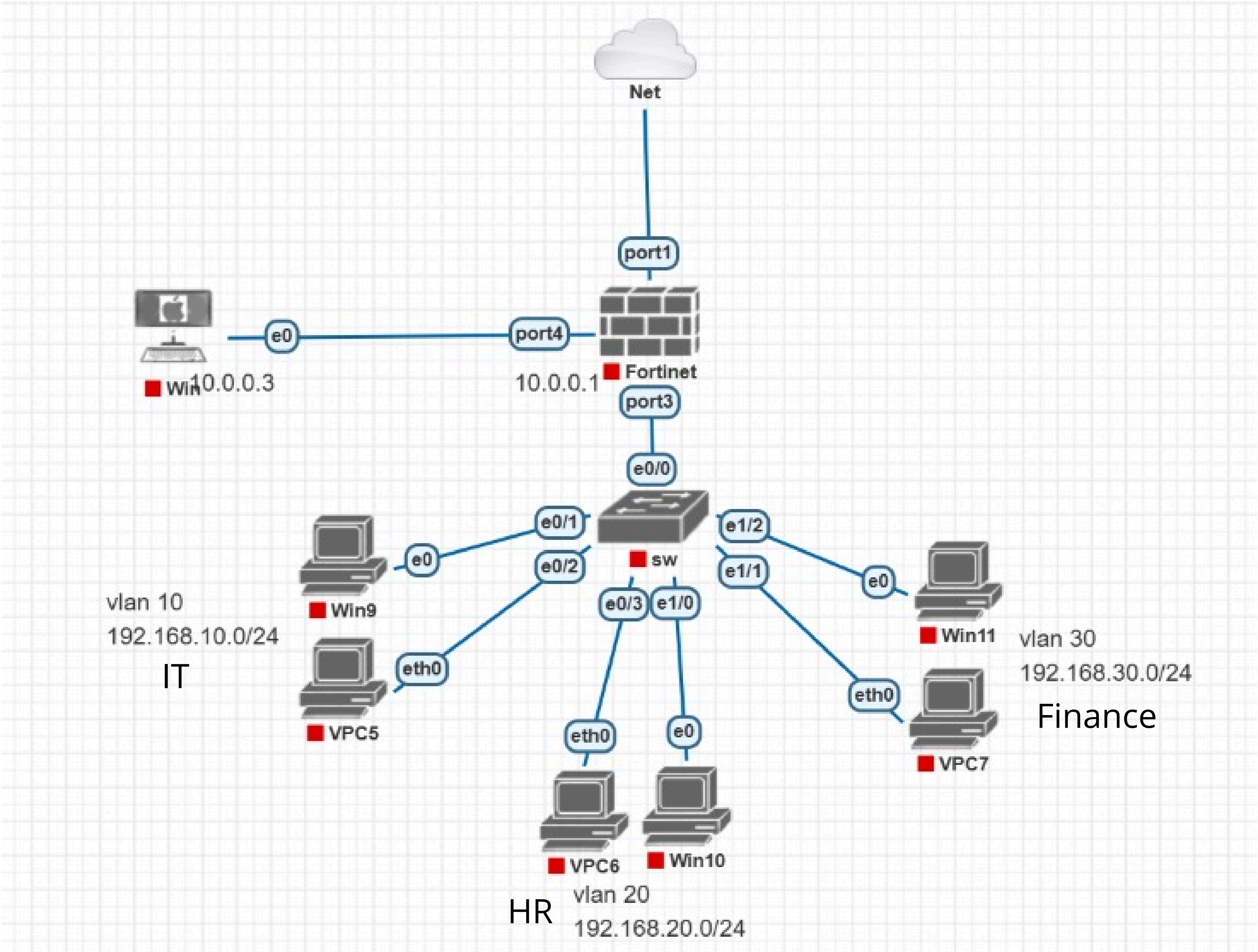
- VLAN 10 → IT Department
- VLAN 20 → HR
- VLAN 30 → Guests

Each VLAN has its own IP range and is isolated from the others.

VLANS

- Benefits of using VLANs
- What is Inter-VLAN Routing

NETWORK TOPOLOGY



Three VLANs are connected to a FortiGate firewall through a Layer 2 switch

VLAN CONFIGURATION ON SWITCH

1. Create vlans on switch and Configured Trunk port

Not secure 192.168.1.9/legacy/topology

sw

```
Switch>enable
Switch#sh
Switch#show vlan br
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et1/3, Et2/0, Et2/1, Et2/2
                                                Et2/3, Et3/0, Et3/1, Et3/2
                                                Et3/3, Et4/0, Et4/1, Et4/2
                                                Et4/3, Et5/0, Et5/1, Et5/2
                                                Et5/3, Et6/0, Et6/1, Et6/2
                                                Et6/3, Et7/0, Et7/1, Et7/2
                                                Et7/3, Et8/0, Et8/1, Et8/2
                                                Et8/3, Et9/0, Et9/1, Et9/2
                                                Et9/3
10   VLAN0010                         active    Et0/1, Et0/2
20   VLAN0020                         active    Et0/3, Et1/0
30   VLAN0030                         active    Et1/1, Et1/2
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
Switch#
```

sw

```
Switch#s
Switch#sh
Switch#show in
Switch#show inte
Switch#show interfaces t
Switch#show interfaces tr
Switch#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Et0/0       on               802.1q         trunking      1

Port        Vlans allowed on trunk
Et0/0       1,10,20,30

Port        Vlans allowed and active in management domain
Et0/0       1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Et0/0       1,10,20,30
Switch#
```

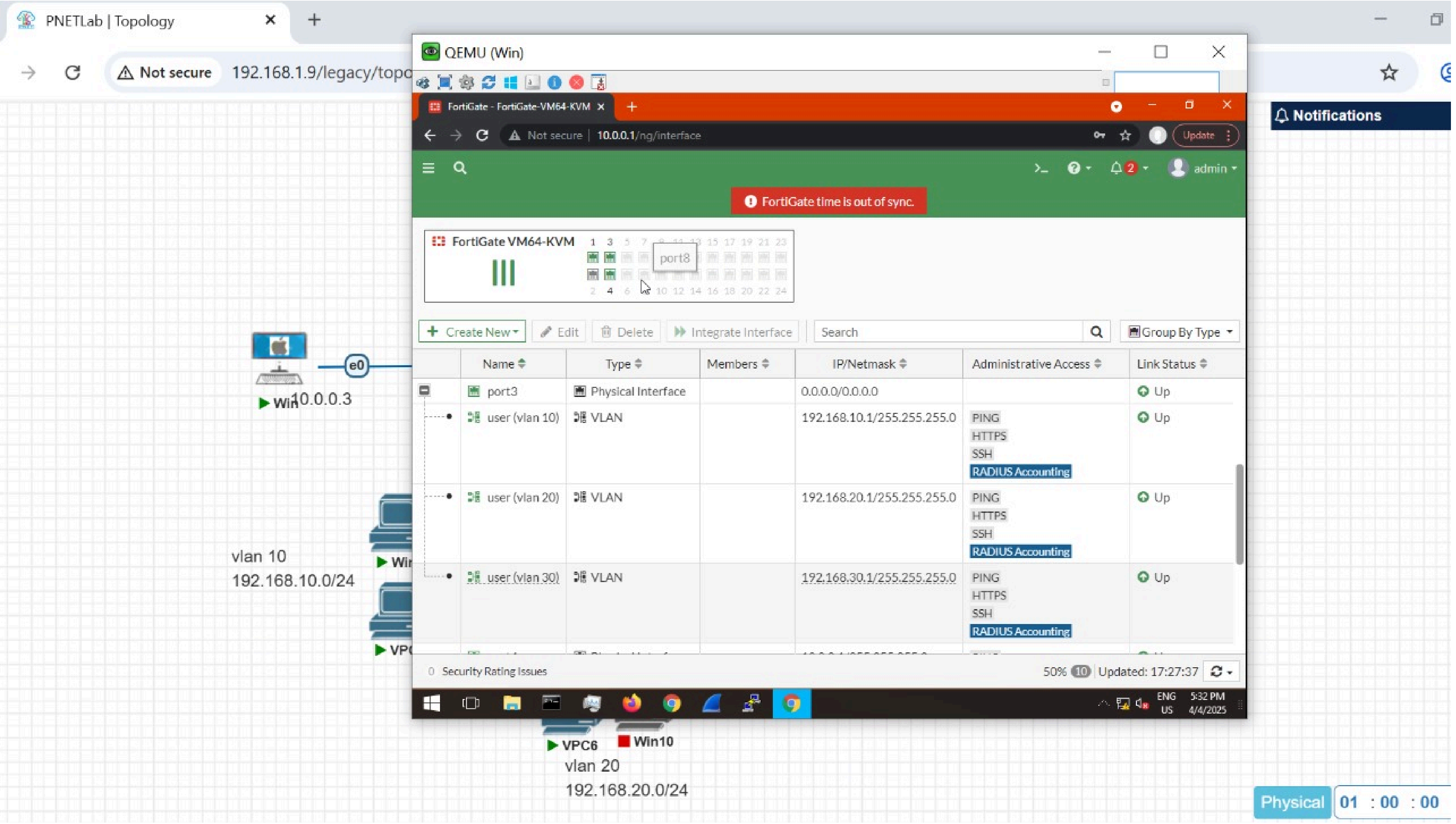
VLANs were created and ports were assigned to their respective VLANs, as shown in the command output.

FORTIGATE CONFIGURATION STEPS

- Create VLAN Interfaces: Go to Network → Interfaces → Create New (VLAN) Assign VLAN IDs (10, 20, 30)
- Configure IP Addressing: Set IP for each VLAN (e.g., 192.168.10.1/24 for VLAN 10
- Enable DHCP (Optional): Assign DHCP ranges for each VLAN
- Firewall Policies: Allow traffic between VLANs as needed (e.g., HR → Finance, but block Guest)

FORTIGATE CONFIGURATION STEPS

- Created subinterface for each VLANs on FortiGate on port 3



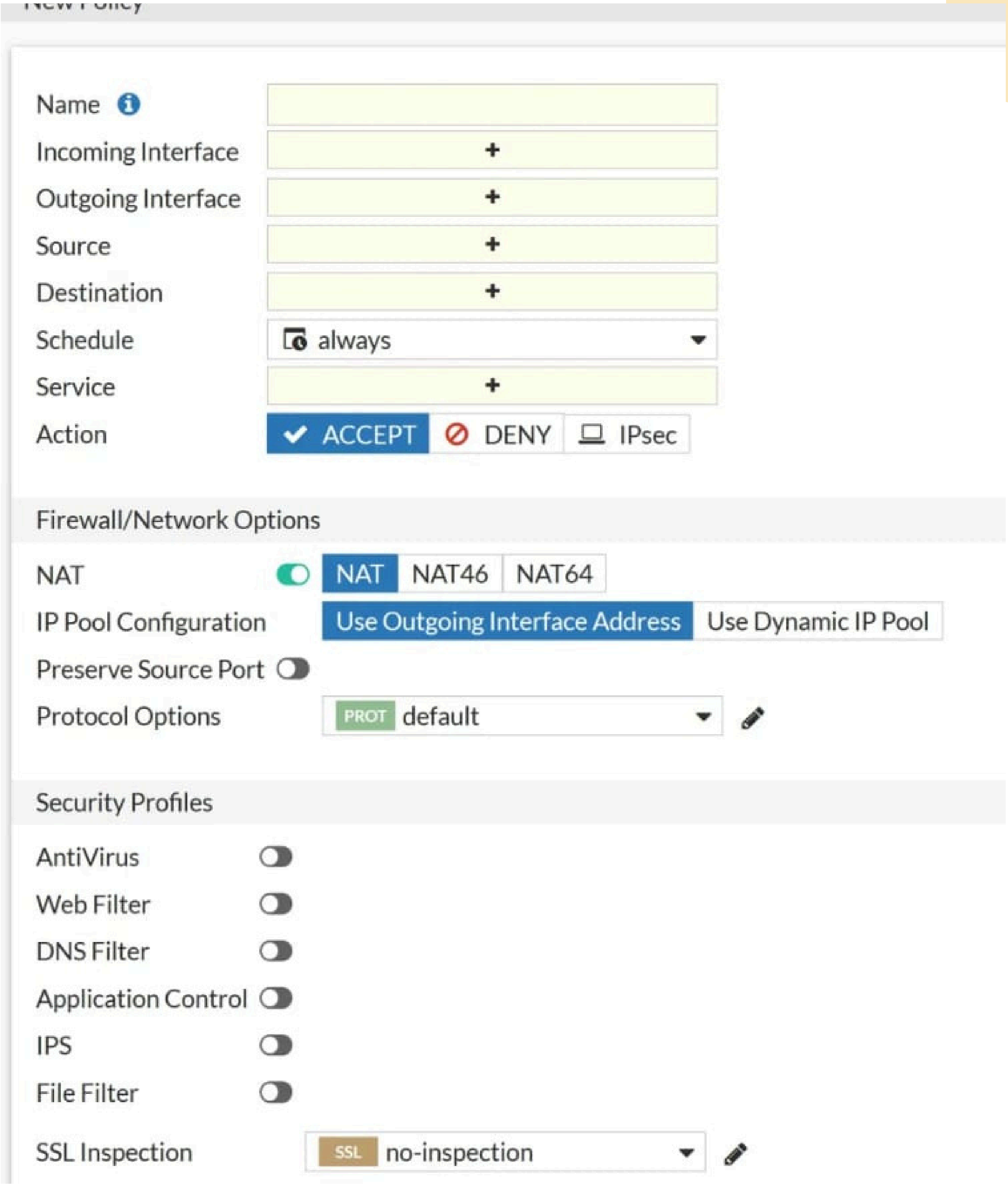
FortiGate policy

What is a Firewall Policy?
A set of rules used to control traffic flow between different networks or VLANs based on criteria like source, destination, service, and action.

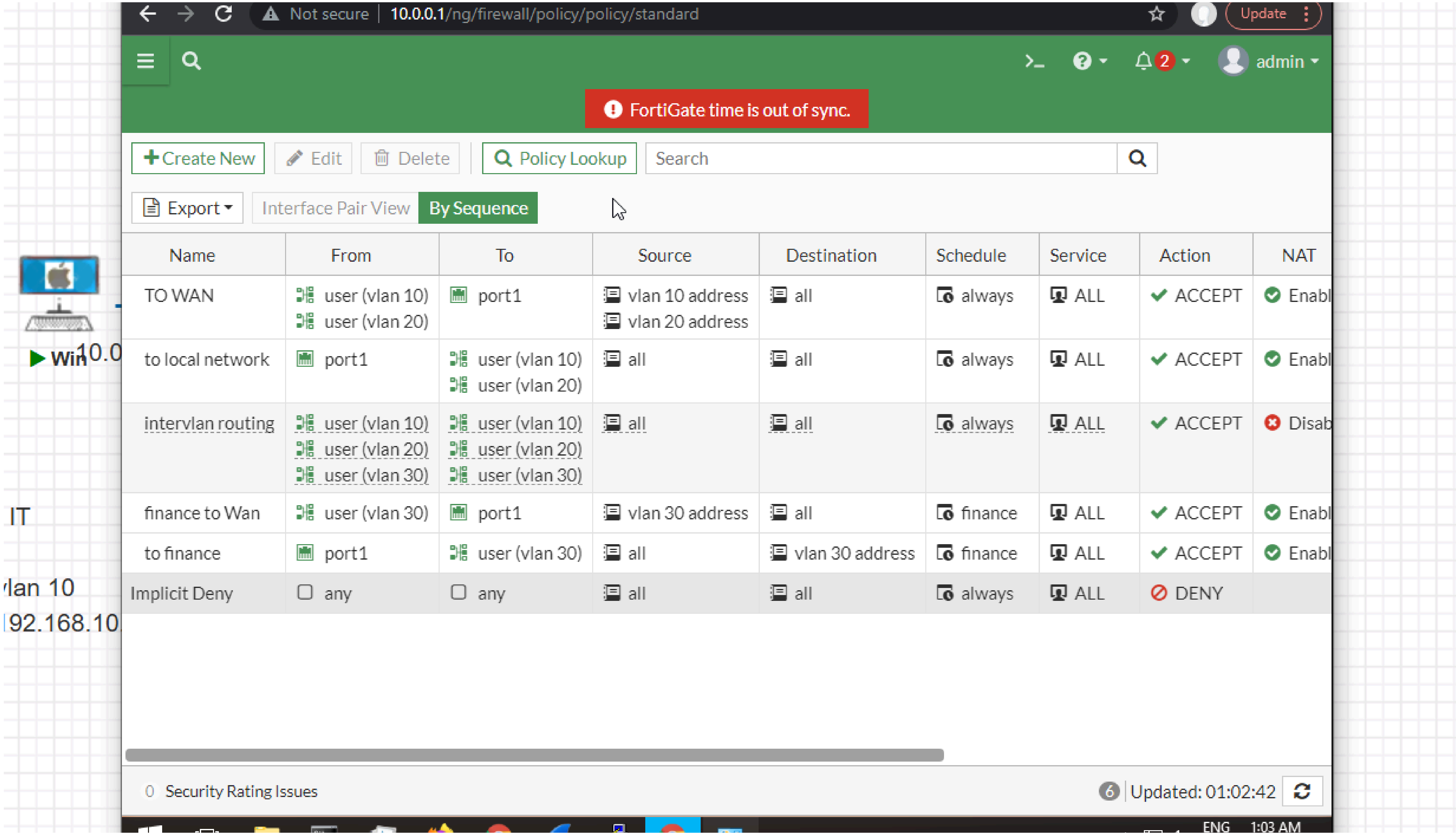
Purpose in Our Project:

• Control communication between VLANs
• Restrict unauthorized acces
• Allow only specific traffic (e.g., HTTP, DNS)

Enable internet access for certain VLANs



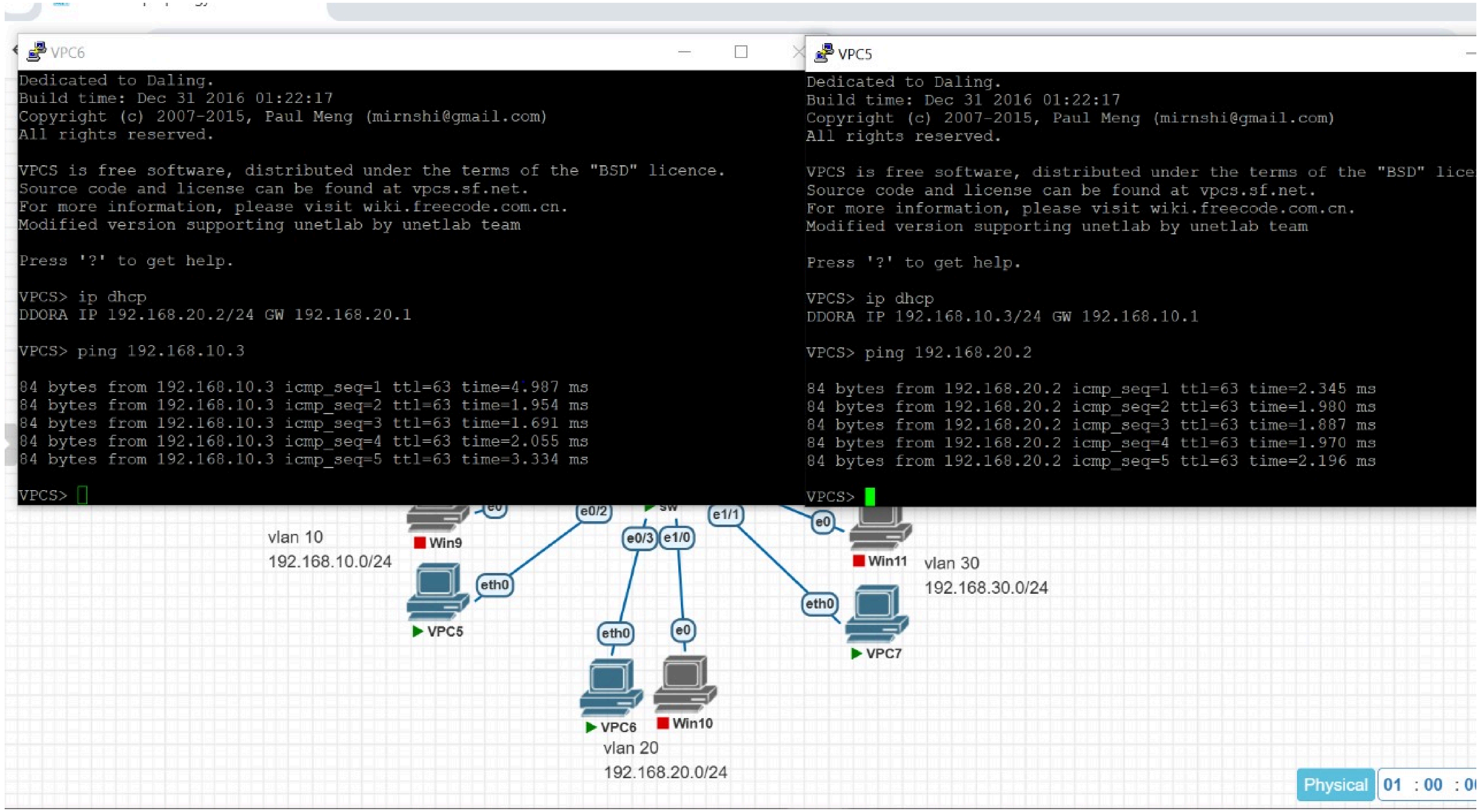
FORTIGATE POLICY



Name	From	To	Source	Destination	Schedule	Service	Action	NAT
TO WAN	user (vlan 10) user (vlan 20)	port1	vlan 10 address vlan 20 address	all	always	ALL	ACCEPT	Enabl
to local network	port1	user (vlan 10) user (vlan 20)	all	all	always	ALL	ACCEPT	Enabl
intervlan routing	user (vlan 10) user (vlan 20) user (vlan 30)	user (vlan 10) user (vlan 20) user (vlan 30)	all	all	always	ALL	ACCEPT	Disab
finance to Wan	user (vlan 30)	port1	vlan 30 address	all	finance	ALL	ACCEPT	Enabl
to finance	port1	user (vlan 30)	all	vlan 30 address	finance	ALL	ACCEPT	Enabl
Implicit Deny	any	any	all	all	always	ALL	DENY	

VERIFICATION & TESTING

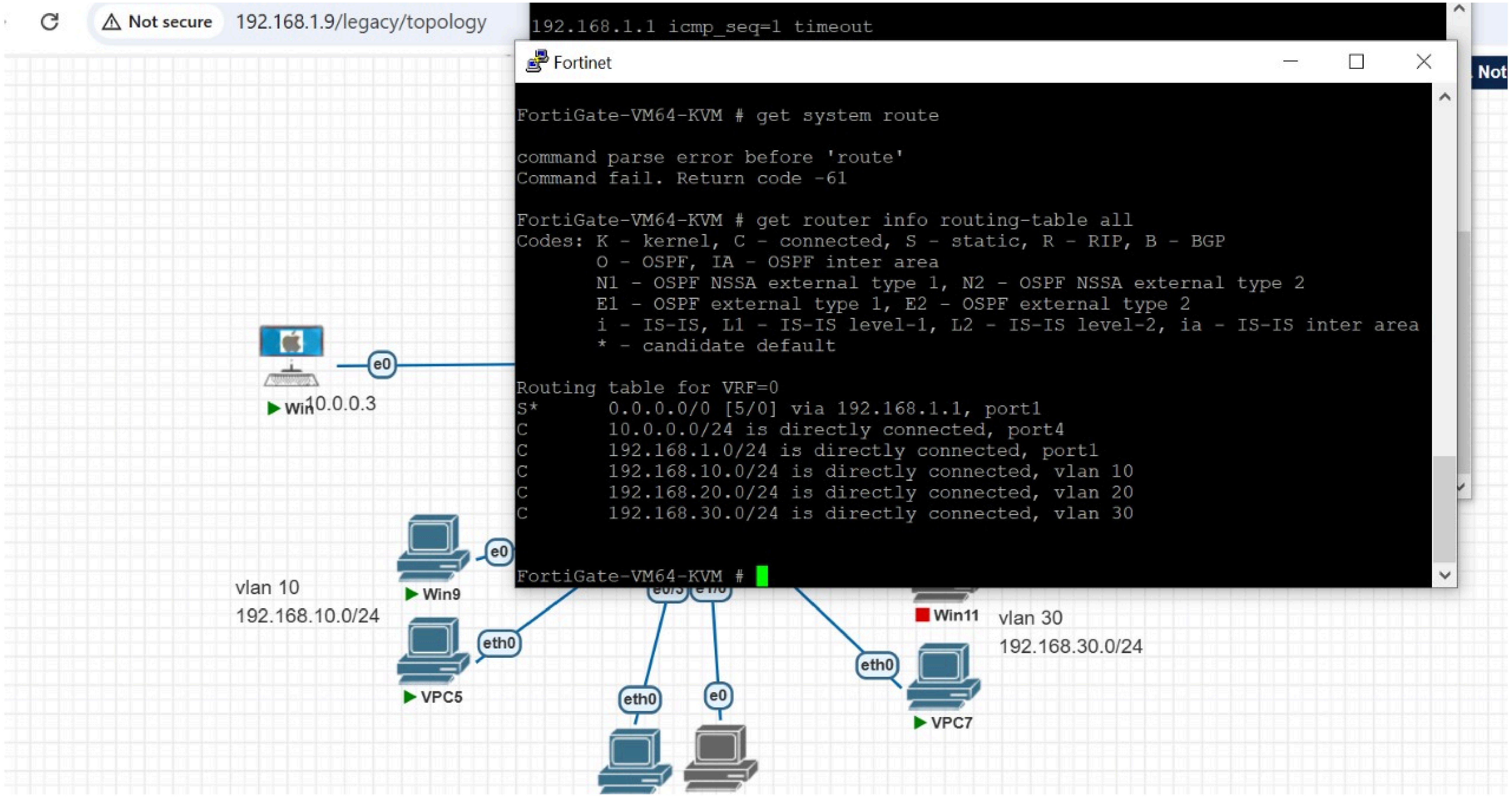
- Ping Test: Verify communication between VLANs



Ping from VLAN 10 to VLAN 20

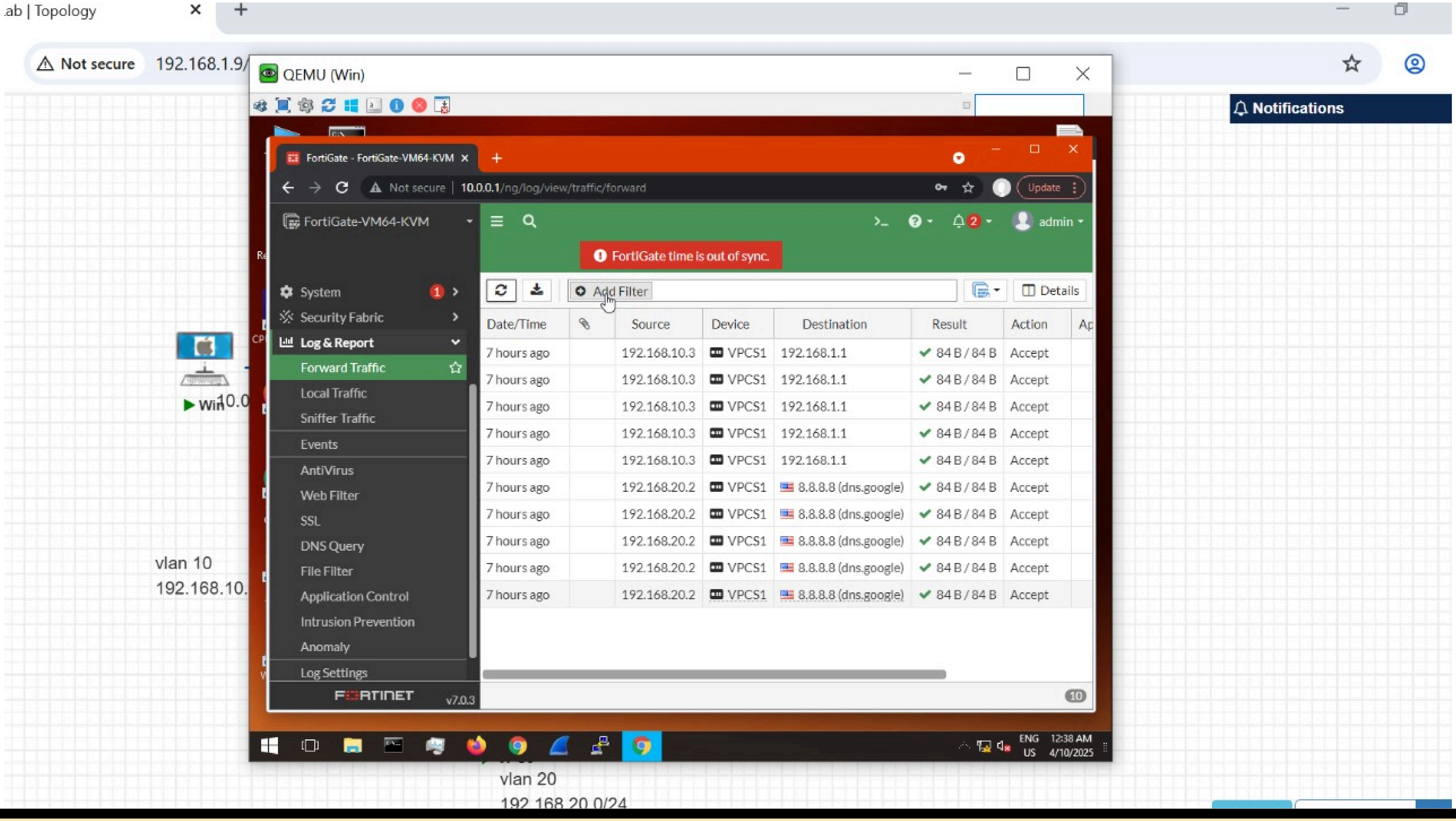
VERIFICATION & TESTING

- Check Routing Table: get router info routing-table all



VERIFICATION & TESTING

- Logs:Monitor traffic in FortiGate logs



CONCLUSION

- VLANs created successfully
- Inter-VLAN routing via FortiGate
- Network segmentation and security improved

THANK YOU